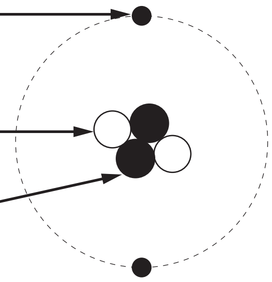
Examiner only

3. A chemistry student correctly draws and labels a helium atom.



(a) An alpha particle is a helium nucleus. State the difference between the structure of an alpha particle and a helium atom. [1]

....

(b) A nucleus of polonium-218 ($^{218}_{84}\mathrm{Po}$) decays by an alpha (α) emission.

$$^{218}_{84}\mathrm{Po} \longrightarrow \mathrm{X} + {}^{4}_{2}\alpha$$

A short time later the resulting nucleus X decays by a beta (β) emission.

$$\mathrm{X} \longrightarrow \mathrm{Y} + {}^{0}_{-1}\beta$$

Nucleus Y then decays by a beta (β) emission.

$$\mathrm{Y} \longrightarrow {}^{214}_{84}\mathrm{Z} + {}^{0}_{-1}\beta$$

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The following table gives information about some elements, their symbols and proton numbers.

Element	Symbol	Proton number
radium	Ra	88
francium	Fr	87
radon	Rn	86
astatine	At	85
polonium	Po	84
bismuth	Bi	83
lead	Pb	82

(i) Use the information above to complete this table. [4]

	Element	Nucleon number
X	....	....
Y	....	....

(ii) State which of X, Y or Z is an isotope of polonium (Po). .... [1]

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